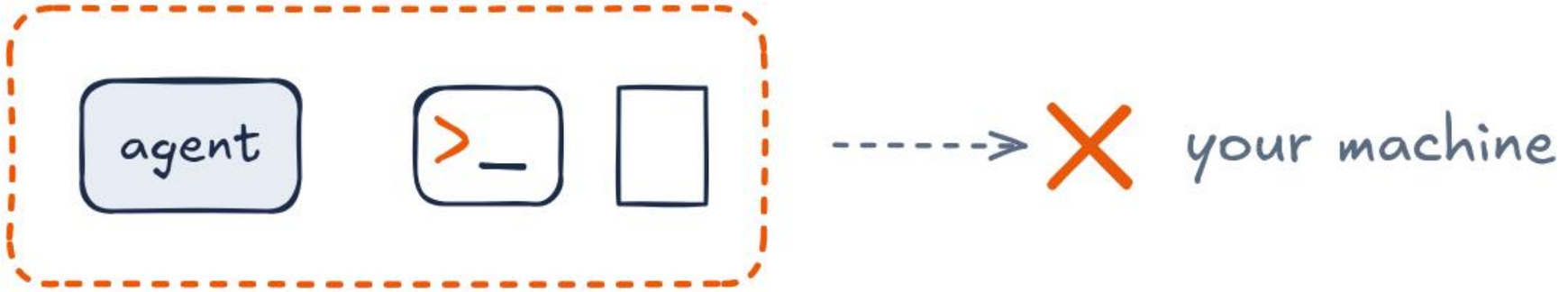
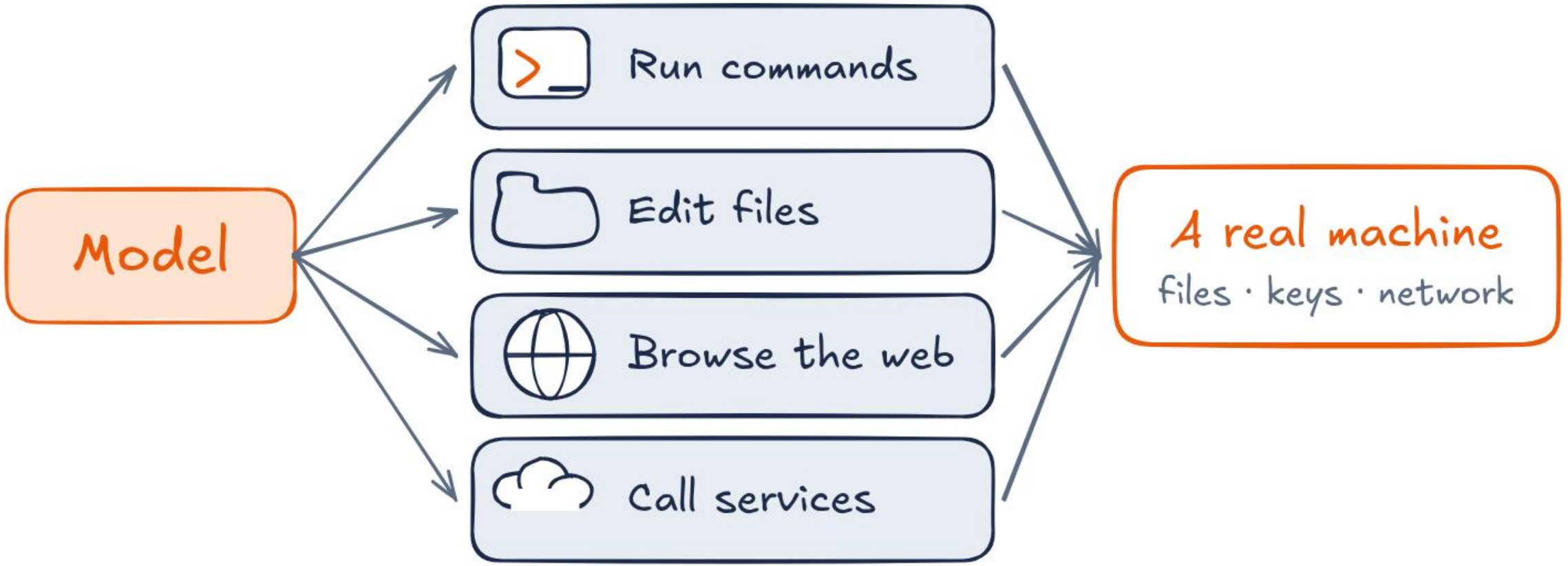
Internal talk · 30 min

Sandboxes for AI agents

Safe places to run code and use a screen



Agents do not only talk



Every tool call has real effects

When an agent gets it wrong

Replit · July 2025

reported by The Register, Business Insider



Code freeze

no changes allowed

Live database deleted

by the agent

"Rollback is impossible"

the agent said

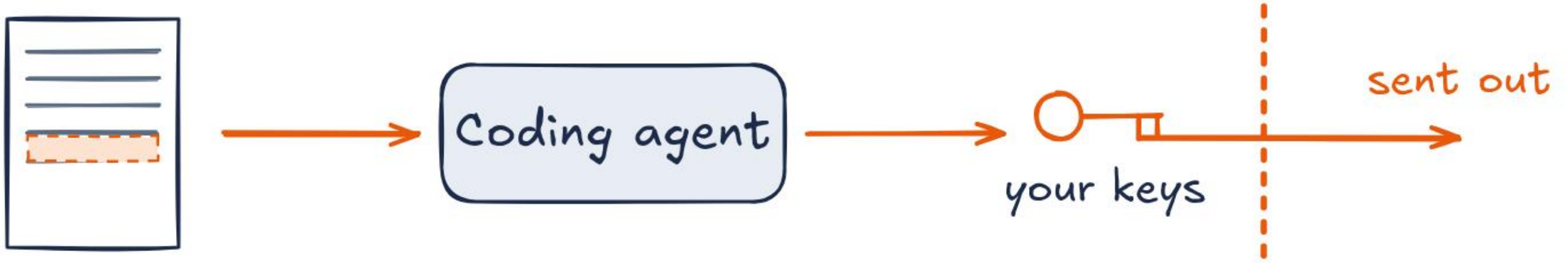
Rollback worked

data restored

Infrastructure saved the data, not the prompt

Hidden instructions

HiddenLayer 2025 · arXiv 2509.22040

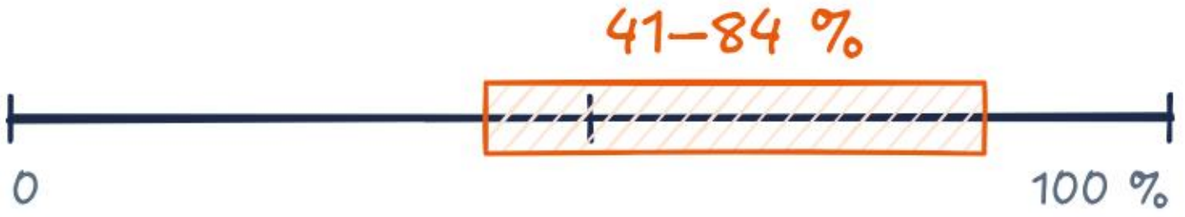


a README

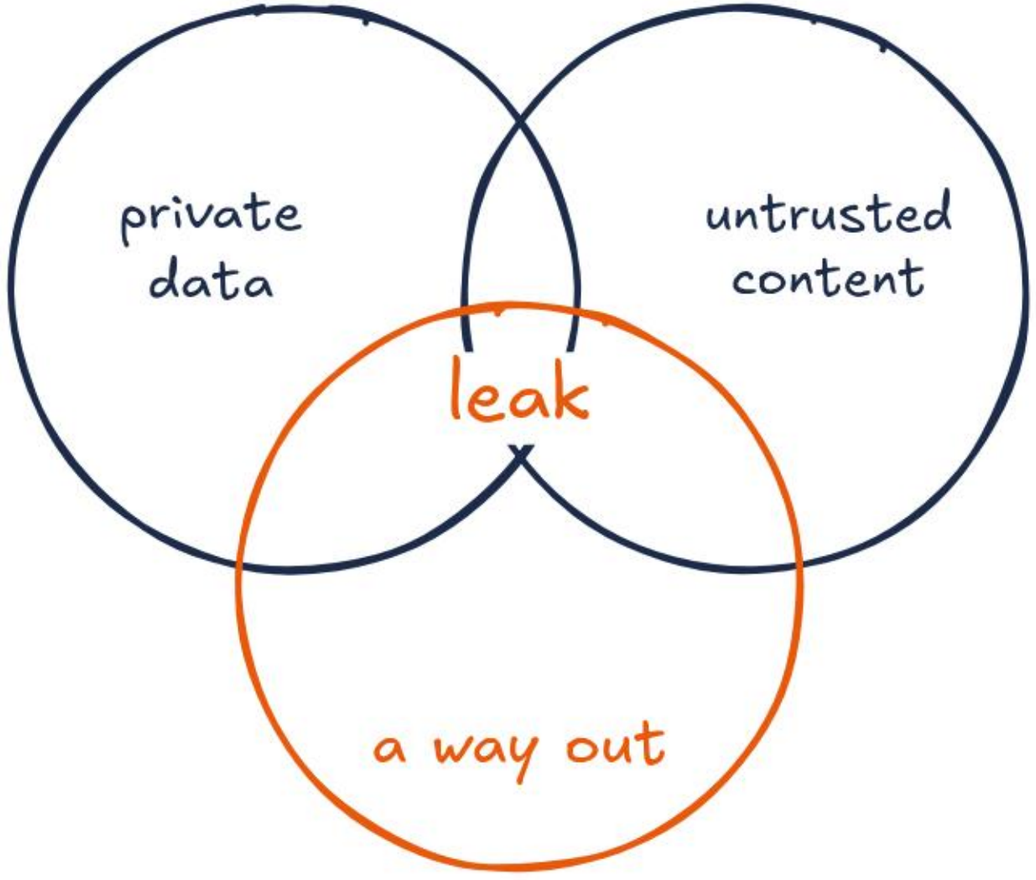
one hidden line

Research study

attack success on coding assistants



Three ingredients of a leak



A sandbox cuts the legs

No real secrets inside

cuts the data

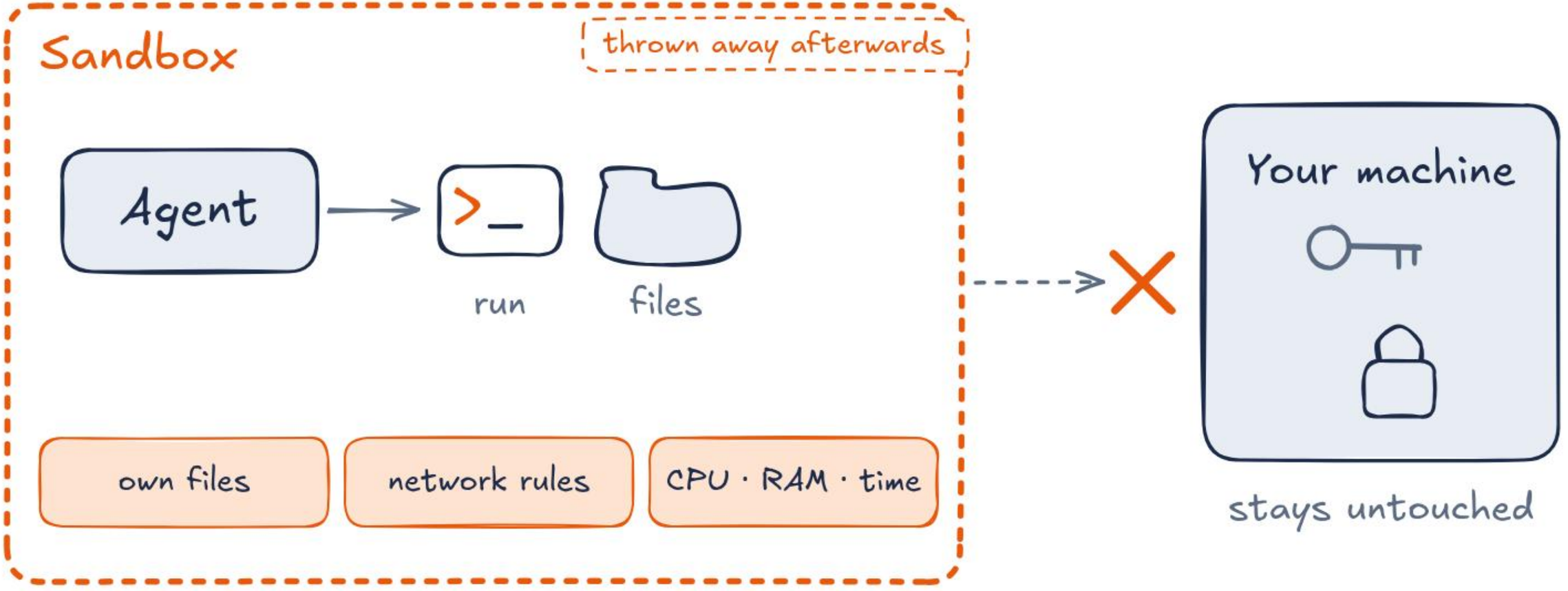
Network off or allowlisted

cuts the way out

Take one away and an attack becomes an annoyance

What is a sandbox?

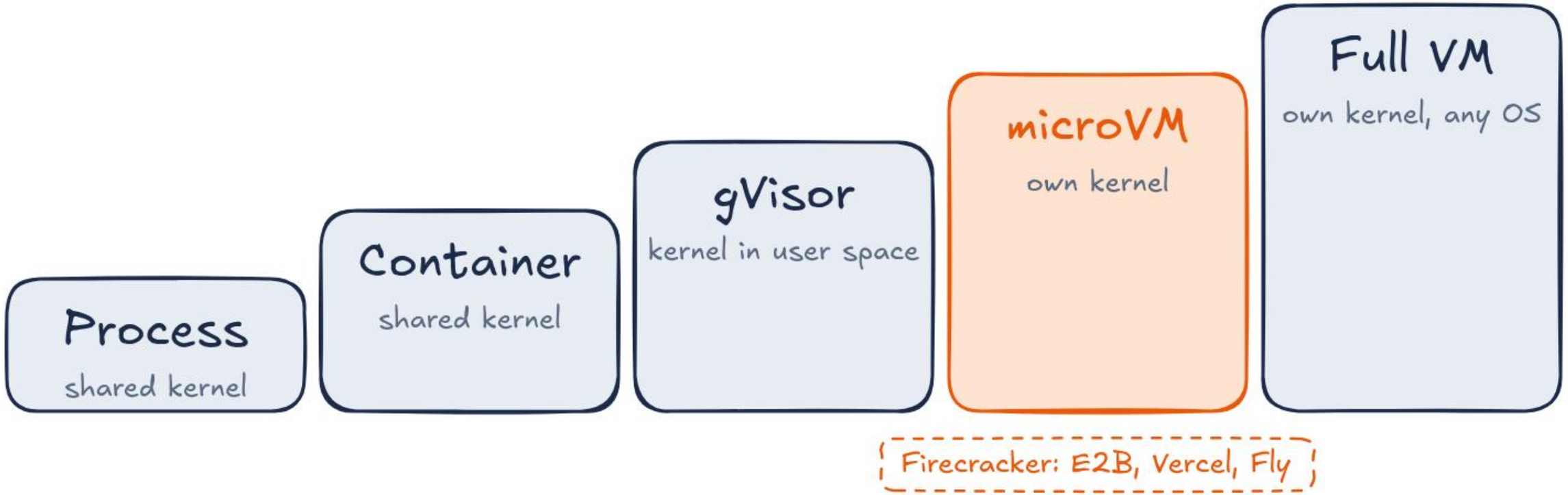
A disposable, walled-off computer for the agent



The isolation ladder

weaker wall → stronger wall

WASM and V8 isolates are a different, lighter kind of sandbox

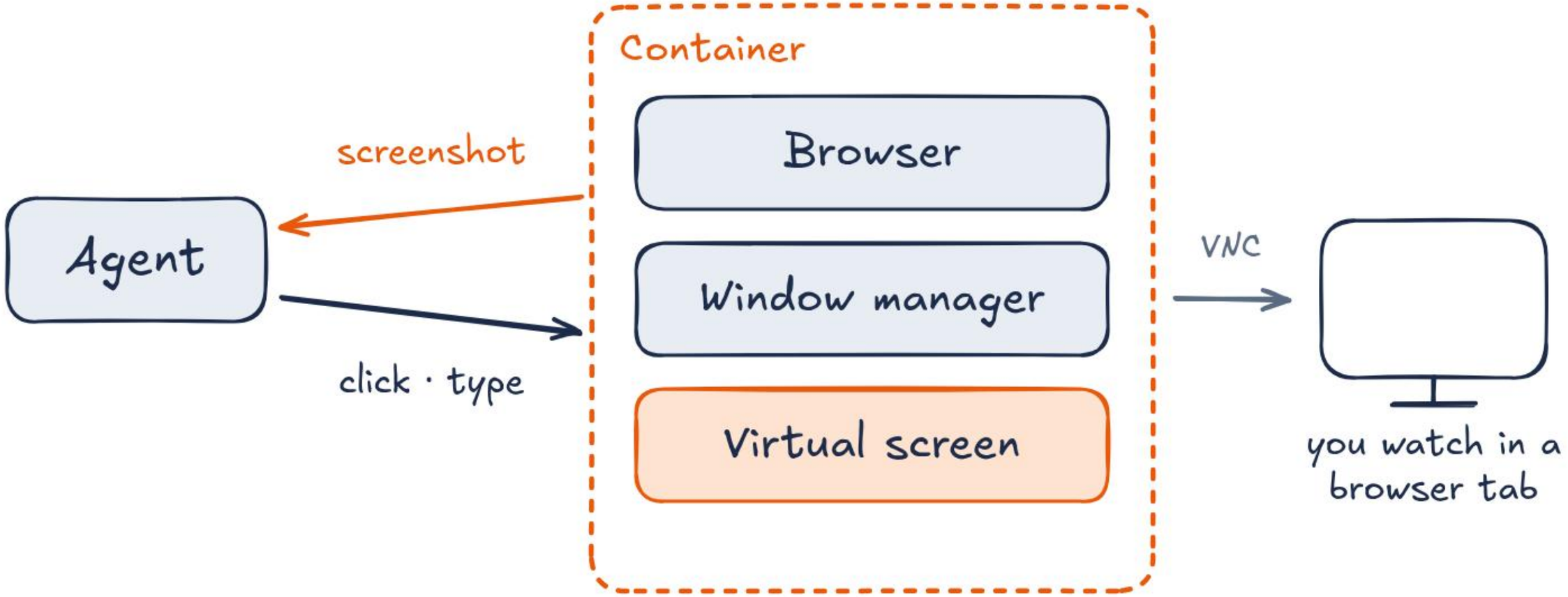


Pick by trust

rule of thumb

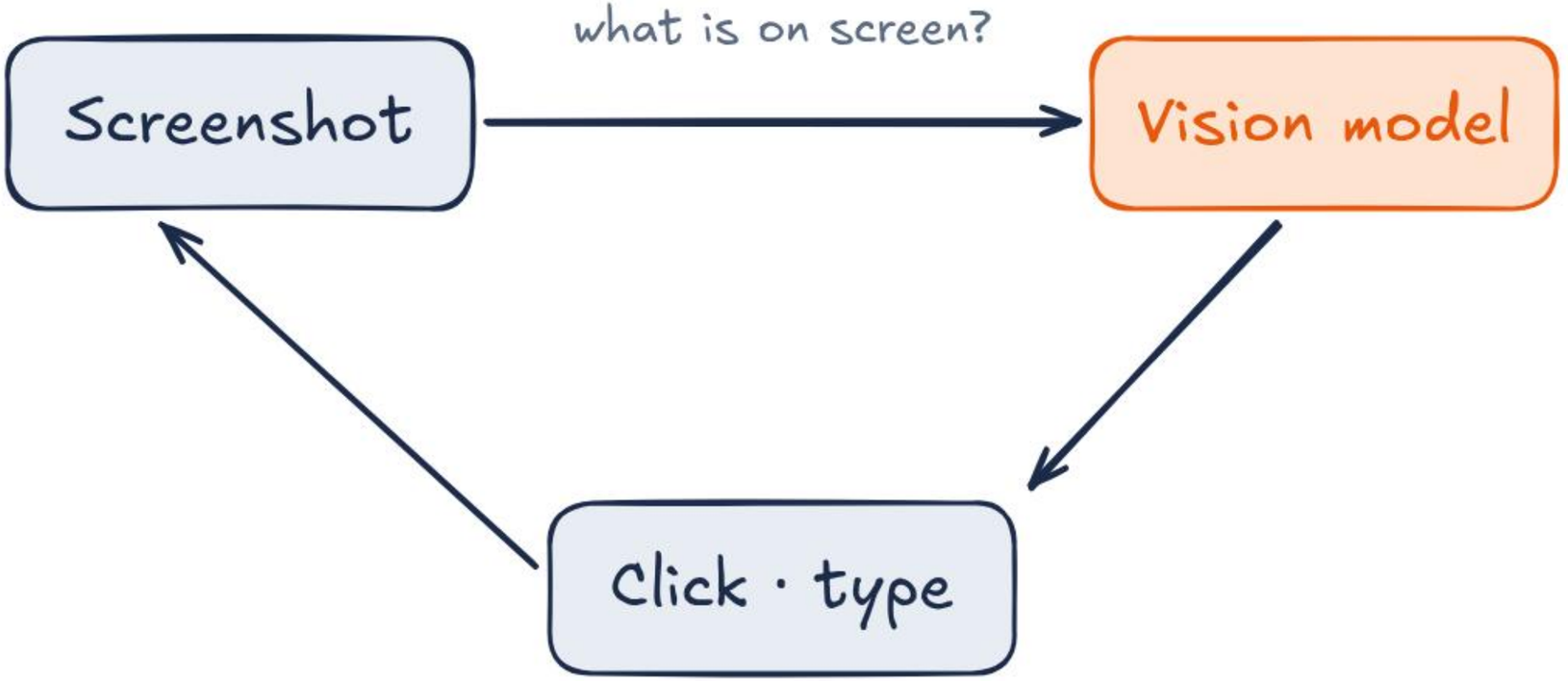
Your own code	→	Container
Code from an LLM, many tenants	→	gVisor or microVM
Needs a screen or Windows	→	Full VM

A screen for the agent



See, think, act

our workshop: ex4



repeat until the model says "done"

Same job, different walls

checked 10 Oct 2026

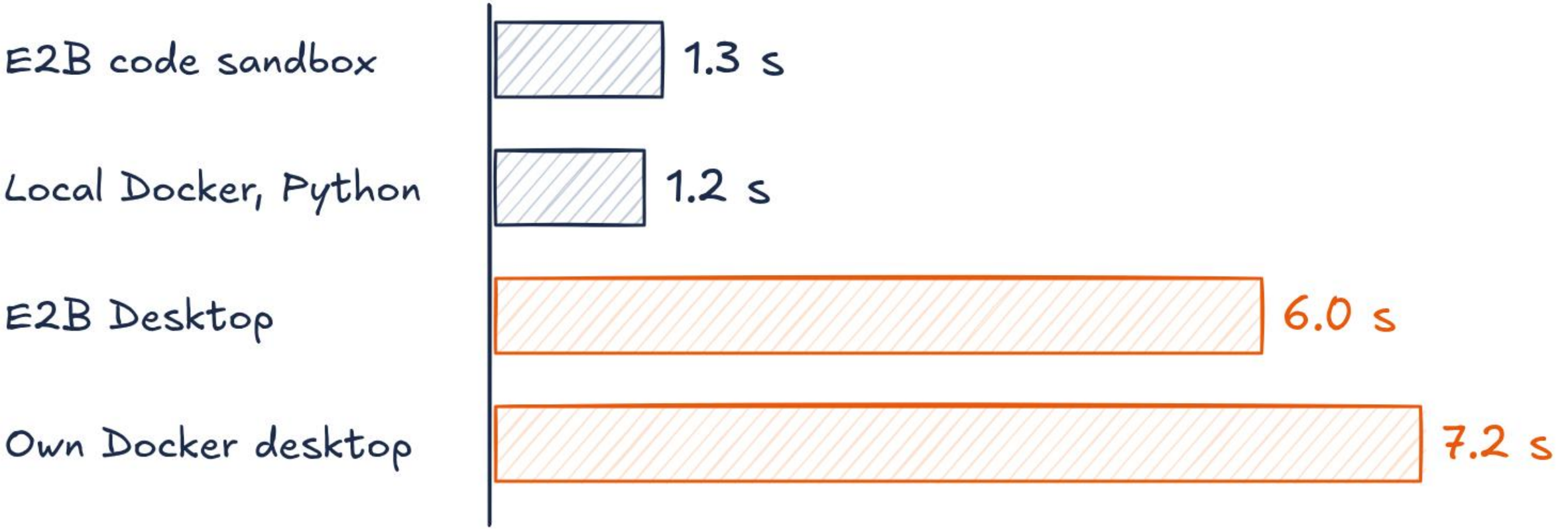
E2B microVM	**Vercel Sandbox** microVM	**Fly.io Sprites** microVM
Cloudflare microVM per container	**Modal** gVisor, VMs since Oct 2026	**Daytona** containers + a VM class

Also: browser clouds, and your own Docker

Measure the cold start

seconds until the sandbox ran its first command

our measurement · 5 runs · median



Vendors quote about 150 ms. Boot time is not first command.

Live demo: the setup

```
docker run -d --rm \
  --network none \
  --memory 512m --cpus 1 \
  --pids-limit 128 \
  --read-only --tmpfs /tmp \
  --user 65534:65534 \
  python:3.12-slim sleep 600
```

→ no network

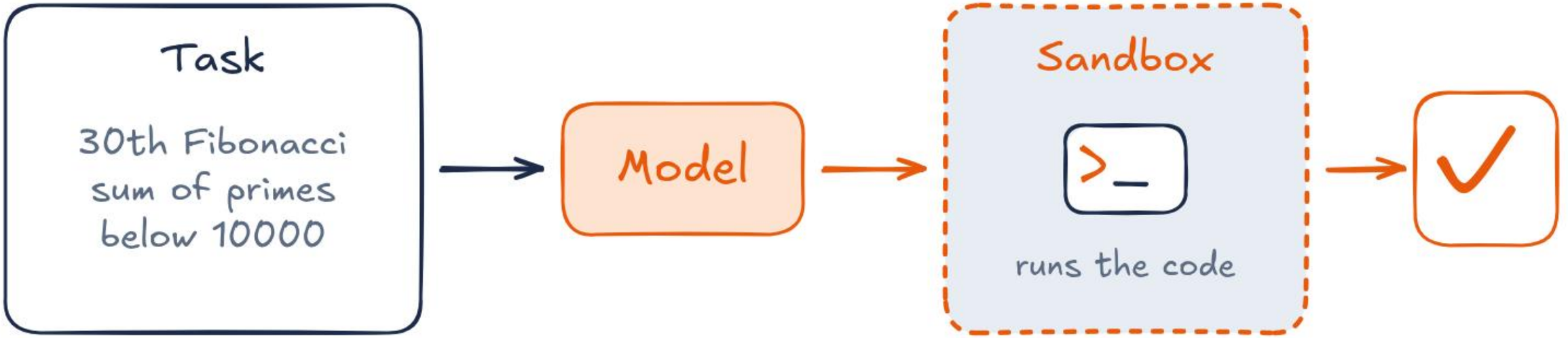
→ capped memory, CPU

→ read-only, not root

python ex2_llm_loop/agent.py "30th Fibonacci number and the sum of primes below 10000"

shortened from ex2_llm_loop/agent.py

Live demo: the result

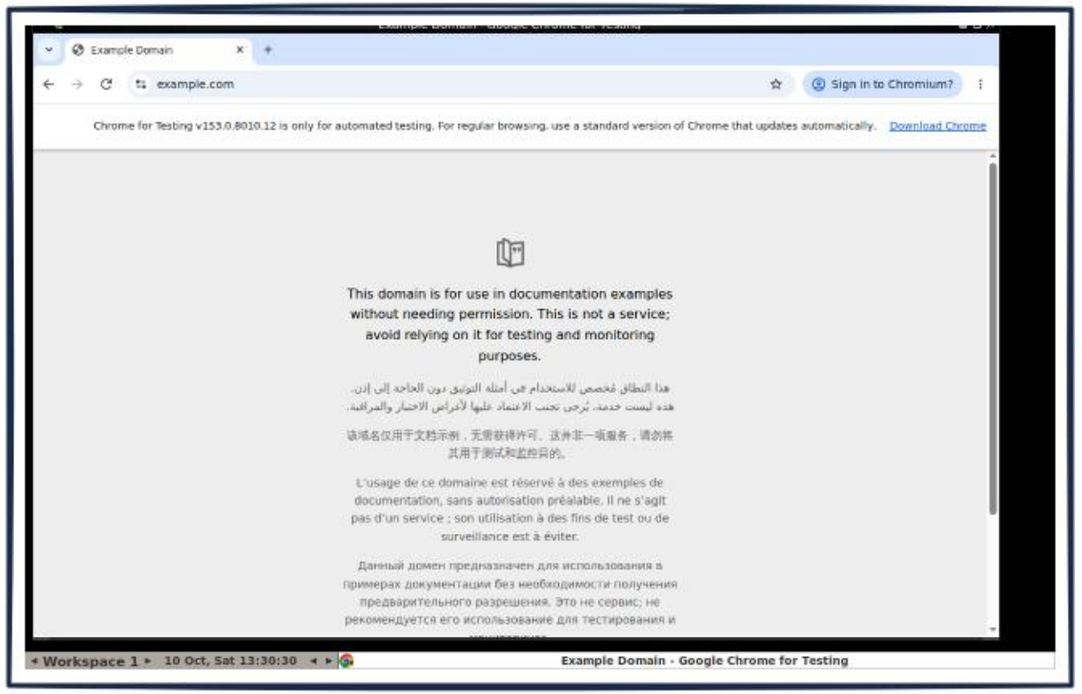


832040 **5736396**

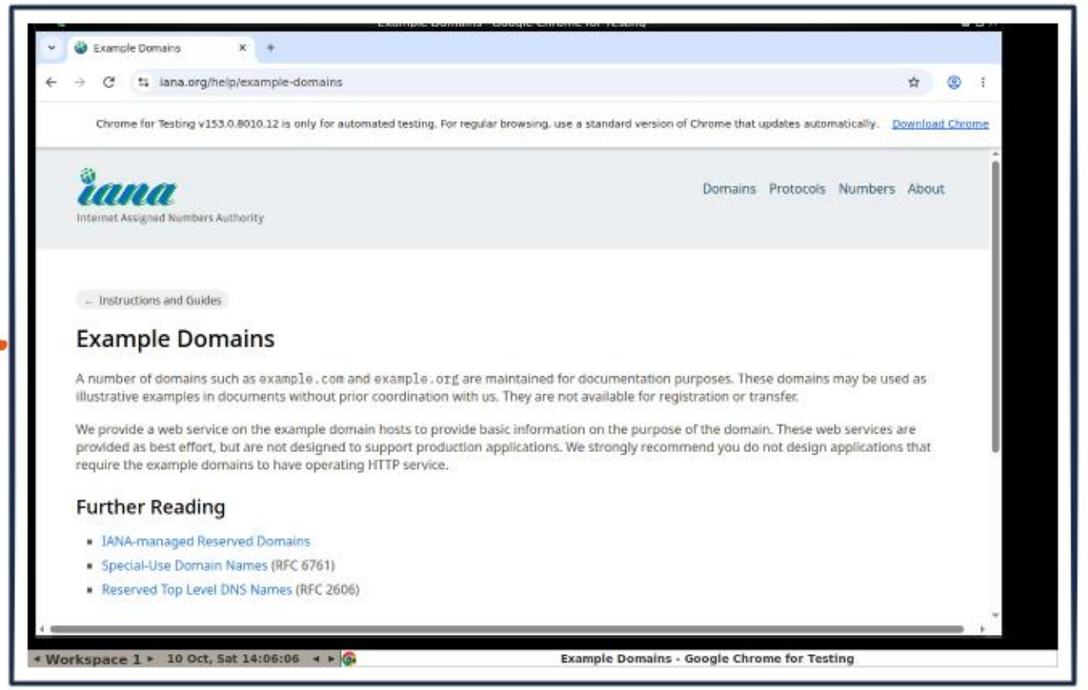
both correct · one tool call · about 12 s

recorded run · free model · local Docker, not E2B

Live demo: a desktop agent



start: example.com



agent ended on iana.org

read the page: 4 steps, about 20 s

click a link: 24 steps, about 6 min

recorded run · free vision model · our own Docker desktop

Honest limits

Free models are flaky

1 of 5 usable on the desktop

Vendor numbers

are claims, not proof

Some details unverified

e.g. isolation of Devin, Cursor agents

Providers move fast

Daytona open source archived

Production checklist

- ✓ Network off or allowlisted
- ✓ No production secrets
- ✓ CPU, memory and time limits
- ✓ Log every command
- ✓ Human OK for destructive steps
- ✓ Backups out of its reach

Three takeaways

1

The prompt is not a wall

2

Match the wall to the trust

3

Measure it, not the brochure

Try it yourself

https://github.com/vusatui/rnd-research
→ sandboxes-and-remote-desktops/

- Run the workshop
- Benchmark a cold start
- Compare the providers
- Read the sources



scan me

Questions?

Sandboxes for AI agents